Umm Al-Qura University
College of Computing
Computer Science and AI Department

Data management and – CS2051
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Data Management Project
(Report)

Team members' names: Yousra Tamer, Reham khan, Afrah Al-Kabeer.
Project Title: Data Management Project.

Supervised by: Dr. Hanan Saad Alshanbari

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1. Introduction

In this project, we collected and analyzed some data about accidents to identify patterns and factors influencing damage costs. The primary objective of this project is to develop a machine learning model that can estimate accident damage costs based on various accident-related features, like: This can help authorities and insurance companies make more data-driven decisions.

2. Project Objectives:

- Identify key factors that influence accident costs.
- Improve data quality through cleaning and preprocessing techniques.
- Develop an accurate machine learning model for damage cost estimation.
- Evaluate model performance using appropriate metrics.

3. Data Collection and Preprocessing:

a. Data Cleaning Process

1. Initial Data Exploration:
   - The dataset originally contained 20,008 rows and 24 columns.

```
[ ] Accidents.shape # Exploring the data, it has 20008 rows and 24 columns
(20008, 24)
```

   - After removing duplicate rows, the dataset was reduced to 20,001 rows.

```
[ ] Accidents.shape
(20001, 15)
```

2. Feature Selection:
   - We dropped unimportant and repeated features to focus on relevant information.

3. Handling Missing Values:
   - Missing values were found only in the total_injuries column.

```
Accidents.isna().sum() # The number of missing values
```

	0
weather_condition	0
lighting_condition	0
first_crash_type	0
alignment	0
roadway_surface_cond	0
road_defect	0
intersection_related_i	0
damage	0
prim_contributory_cause	0
num_units	0
most_severe_injury	0
injuries_total	3
crash_hour	0
crash_day_of_week	0
crash_month	0

- We filled these missing values using the median imputation to maintain data consistency.

```
[ ] Accidents['injuries_total'].unique() # before filling the missing values

array([ 0.,  1., nan,  3.,  2.,  7.,  4.,  5.,  6., 10.,  9., 15.,  8.,
       11.])

[ ] imputer = SimpleImputer(missing_values = np.nan,strategy = 'median') #filling the missing values using SimpleImputer()
    Accidents[['injuries_total']] = imputer.fit_transform(Accidents[['injuries_total']])

[ ] Accidents['injuries_total'].unique() # After filling the missing values

array([ 0.,  1.,  3.,  2.,  7.,  4.,  5.,  6., 10.,  9., 15.,  8., 11.])
```

4. Simplifying “prim_contributory_cause” column:
   - This column had too many unique values, making it difficult to process.
   - We selected the six most frequent values and assigned all other values to a new category called "other" to reduce complexity and improve model performance.

b. Data transformation

After cleaning the data, we proceeded with Transformation to ensure that all categorical features converted to numerical and that numerical features are in the same range. This is particularly important for models like KNN, which rely on distance-based calculations, and for ensuring stability in Logistic Regression training.

- Converting Categorical Variables into Numerical Representations:

A new encoded version of the dataset: “Accidents_encoded”, was created.

First, Label encoding was applied to ordinal categorical variables to convert them into numerical representations, making them suitable for machine learning models. The encoding was applied to the following columns:

1. "intersection_related_i" encoding:
   - 'Y' is represented as 1.
   - 'N' is represented as 0.

2. "damage" encoding:
   - 'OVER $1,500' is represented as 2.
   - '$501 - $1,500' is represented as 1.
   - '$500 OR LESS' is represented as 0.

3. "most_severe_injury" encoding:
   - 'FATAL' is represented as 4.
   - 'INCAPACITATING INJURY' is represented as 3.
   - 'NONINCAPACITATING INJURY' is represented as 2.
   - 'REPORTED, NOT EVIDENT' is represented as 1.
   - 'NO INDICATION OF INJURY' is represented as 0.

4. "time_of_day" encoding:
   - 'Evening' is represented as 2.
   - 'Afternoon' is represented as 1.
   - 'Morning' is represented as 0.

Second, One-Hot Encoding was applied to the nominal categorical variables to transform them into binary numerical representations (0/1).

- weather_condition
- lighting_condition
- first_crash_type
- alignment
- roadway_surface_cond
- road_defect
- Top_6_contributory_causes

These encodings were applied to the respective columns in the dataset, resulting in numerical values that can be used for model building.

- Scalling data features:
To ensure that the features are within a similar range and to help the model better understand and analyze the data, standardization was applied to the dataset. The goal of this step is to scale the features so that they all have comparable values, which can improve the performance of many machine learning models.

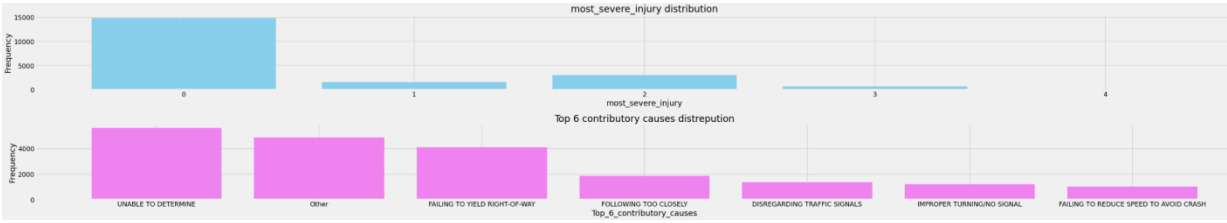
The following columns were selected for scaling:
- 'intersection_related_i'
- 'most_severe_injury'
- 'injuries_total'
- 'crash_hour'
- 'crash_day_of_week'
- 'crash_month'
- 'time_of_day'
- 'num_units'

The StandardScaler() from scikit-learn was used to scale these features. This transforms the data by centering it around zero and scaling it to have a standard deviation of one, ensuring that the features have similar magnitudes and making it easier for the model to process the data efficiently. The scaled data was stored in a new dataset: "Accidents_Scaled".

4. Exploratory Data Analysis Findings:

a. Univariate Analysis:

- Categorical data:

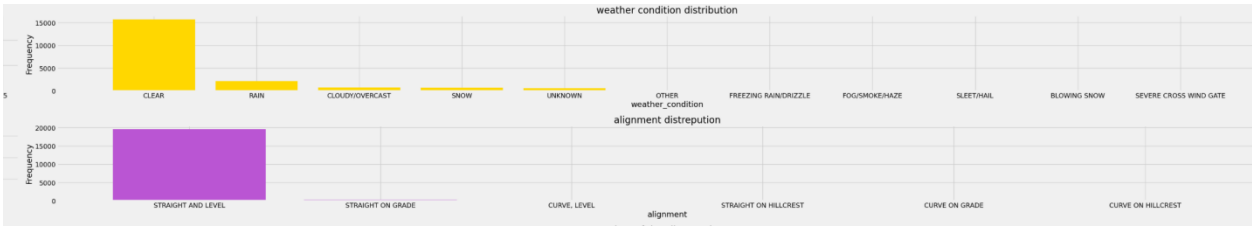


Graph1 (most severe injury): we notice that the most frequently repeted column is '0', which represent the value 'NO INDICATION OF INJUTY' in the actual data which means that most of the accendint cases does not cause injuries.

Graph2 (Top 6 injury causes): we notice that the most common cause of accidents is "FAILING TO YIELD RIGHT-OF-WAY", which refers to not giving the right-of-way or failing to yield to other vehicles according to traffic rules. This occurs when a driver does not adhere to the right-of-way of another vehicle and it is one of the most common causes of accidents globally.

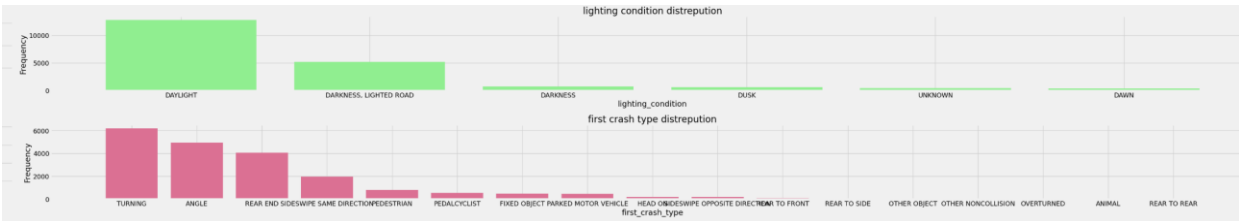
Graph3 (damage): we notice that the most frequently repeted column is '2', which represent the value 'OVER $1,500' in the actual data, which means that most of the accendint damages costs more than 1,500 dollar.

Graph4 (road defect): we notice that the most frequently repeted column is 'NO DEFECTS', mybe that is because the area from which the data was collected has well-maintained roads with no visible defects or irregularities.



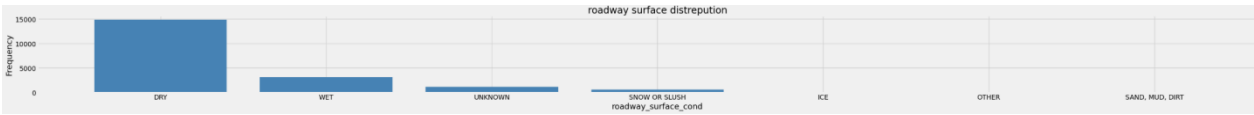
Graph5 (Weather condition): we notice that the most frequently repeated column is 'CLEAR'. Maybe that is because the weather of the area from which the data was collected is typically clear, with stable atmospheric conditions and minimal occurrences of adverse weather.

Graph6 (Alignment): we notice that the most frequently repeted column is 'STRAIGHT AND LEVEL' , this may indicate that accidents often occur in straight roads with clear visibility, possibly due to high speeds or lack of caution. Another possible reason is that the data was collected from an area with mostly flat terrain. Naturally, most recorded accidents would be on Straight and Level roads.



Graph7 (Alignment): we notice that the most frequently repeted column is 'STRAIGHT AND LEVEL' , this may indicate that accidents often occur in straight roads with clear visibility, possibly due to high speeds or lack of caution. Another possible reason is that the data was collected from an area with mostly flat terrain. Naturally, most recorded accidents would be on Straight and Level roads.

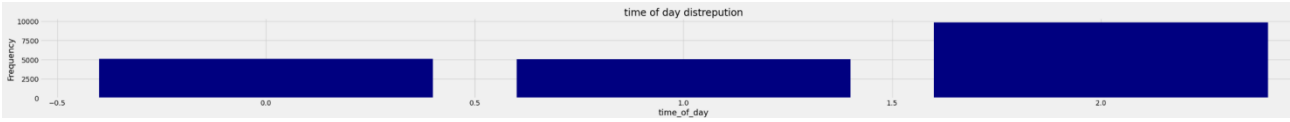
Graph8: Turning collisions are the most damaging because the vehicle may be in a crossing path, increasing the likelihood of a collision with another vehicle approaching at high speed. This is followed by rear-end collisions, which usually occur due to sudden braking or not leaving enough distance, leading to a strong impact, especially at high speeds. Then comes angle collisions, which are also associated with road intersections.



Graph9 (Roadway surface): we notice that Most accidents occur on dry roads because this is the normal condition most of the time. However, when the road becomes wet due to rain, for

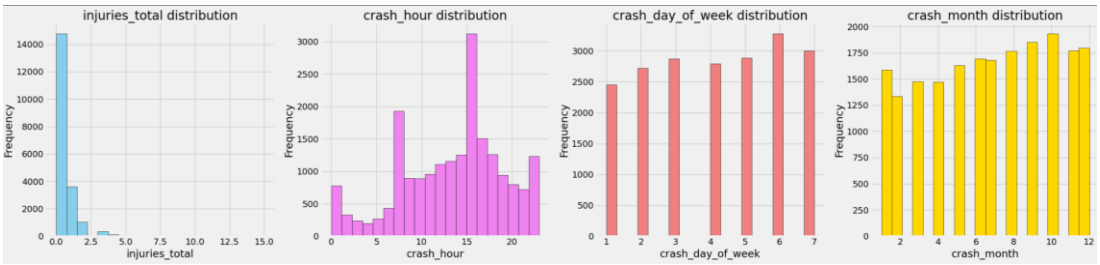
example, the risk of accidents increases due to the sudden decrease in traction, which may lead to loss of vehicle control.

Graph10 (Intersection related): We notice that accidents are more frequent at intersections, which is obvious due to the overlapping lanes and multiple points of vehicle convergence, increasing the likelihood of collisions, especially when speed is misjudged or the right of way is not given. Additionally, running red lights increases the risk of accidents in these areas.

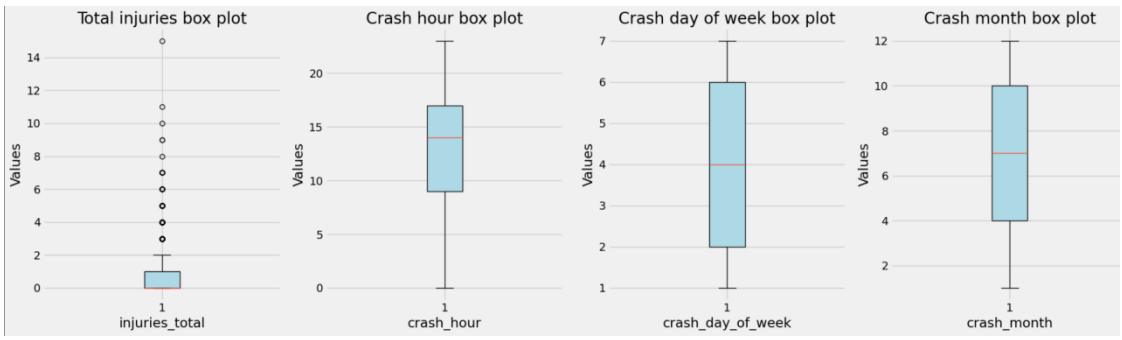


Graph11 (time of day): we notice that the most frequently repeted column is '2', which represent the value 'Evening' in the actual data, that means that accedints happens mostly in the evening.

- Numerical data:



Injuries total distreputon: most accidents injuries in the data are minor.in the data are
Crash hour distreputon: most accidents happen in 3:oo pm
Crash day of week distreputon: most accidents happen on Thursday and Friday.
Crash month distreputon: most accidents happened from August to October.



Total injuries box plot: This plot displays the distribution of injury counts in various accidents, It helps identify the median values (no injuries), the minimum and maximum boundaries (2), as well as the outliers. The presence of multiple outliers indicates rare accidents with very high injury counts.

Crash hour box plot: This plot illustrates how accidents are distributed across different hours of the day.

Crash day of week box plot: This plot shows the distribution of accidents across the days of the week.

Crash month box plot: This plot presents the distribution of accidents across different months of the year.

- The central tendencies

	Mean	Median	Mode	Variance	Standard Deviation
intersection_related_i	0.952102	1.0	1.0	0.045606	0.213555
damage	1.601520	2.0	2.0	0.437706	0.661593
num_units	2.064597	2.0	2.0	0.162427	0.403022
most_severe_injury	0.479826	0.0	0.0	0.758305	0.870807
injuries_total	0.389181	0.0	0.0	0.658631	0.811561
crash_hour	13.383331	14.0	16.0	31.425200	5.605818
crash_day_of_week	4.139143	4.0	6.0	3.922488	1.980527
crash_month	6.792610	7.0	10.0	11.733187	3.425374
time_of_day	1.238088	1.0	2.0	0.691011	0.831271

1. Mean:

The mean represents the average value of each variable, providing an overall sense of the data distribution. However, it can be affected by outliers.
For example, the mean for “Crash_hour” is 13.38, indicating that most accidents occur around 1 PM or close to it.

2. Median:

The median is the middle value when the data is sorted in ascending order. It is more stable than the mean in the presence of outliers as it reflects the true central value of the distribution.
For instance, the median for "Crash_hour" is 14, meaning that half of the accidents happen before 2 PM and the other half after.

3. Mode:

The mode is the most frequently occurring value in a variable. In some cases, there may be more than one mode if multiple values appear with the same frequency.
For example, the mode for "Crash_day_of_week" is 6, indicating that Thursday is the most common day for accidents.

4. Variance:

Variance measures how much the values deviate from the mean. A high variance indicates that the values are widely spread out, while a low variance suggests that they are more concentrated around the mean.
For example, crash_hour has a variance of 31.42, suggesting a wide variation in the hours when accidents occur, and crash_month shows similar variability.

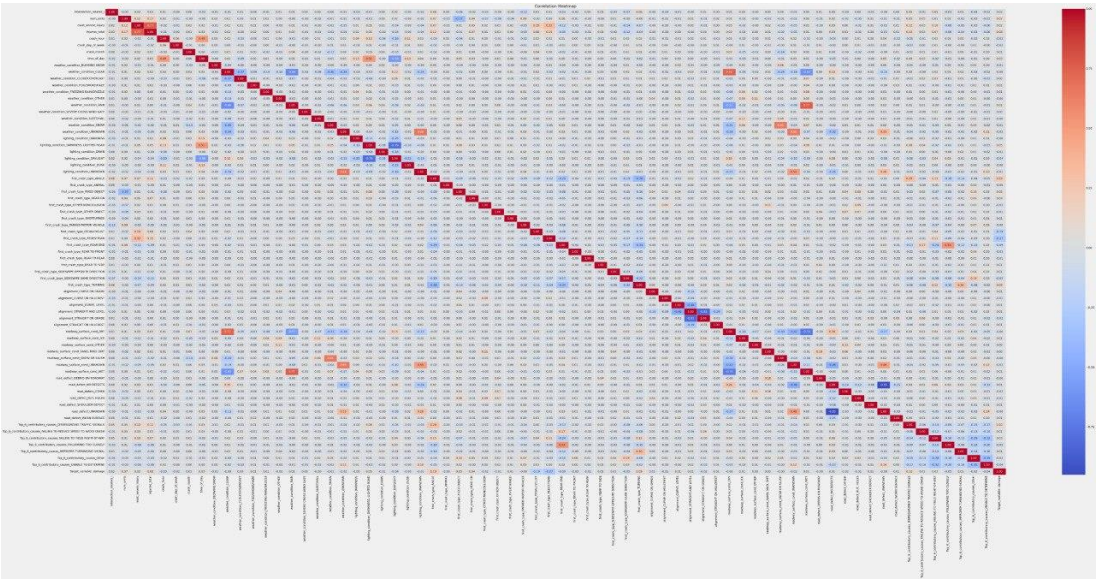
5. Standard Deviation:

The standard deviation is the square root of the variance, representing how spread out the values are around the mean. A high standard deviation means the data points are more dispersed, while a low standard deviation means they are closer together.
For instance, "injuries_total" has a standard deviation of 0.81, indicating that the number of injuries in accidents does not vary significantly between cases.

b. Bivariate/Multivariate Analysis:

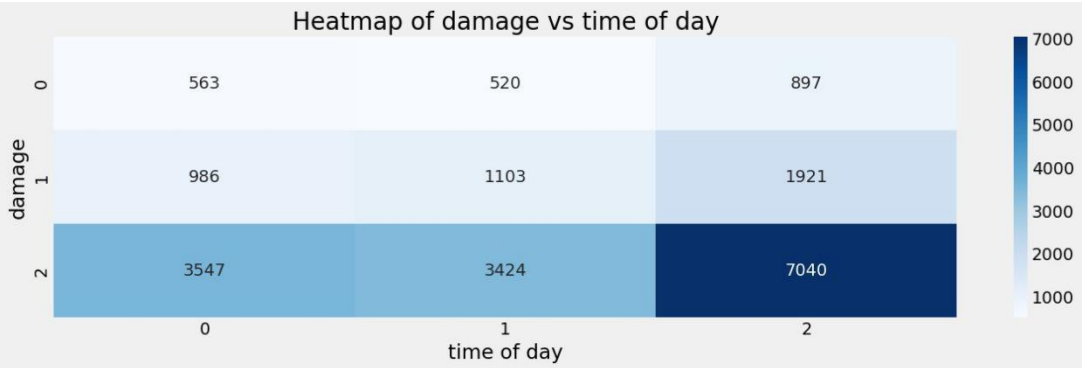
We used heatmaps, first, we calculated the correlation matrix for features to identify the most correlated features with the target variable. After visualizing the heatmap, we can identify the features most correlated with the target variable and we used it with the partial heatmaps to clarify it.

- The comprehensive heatmap:



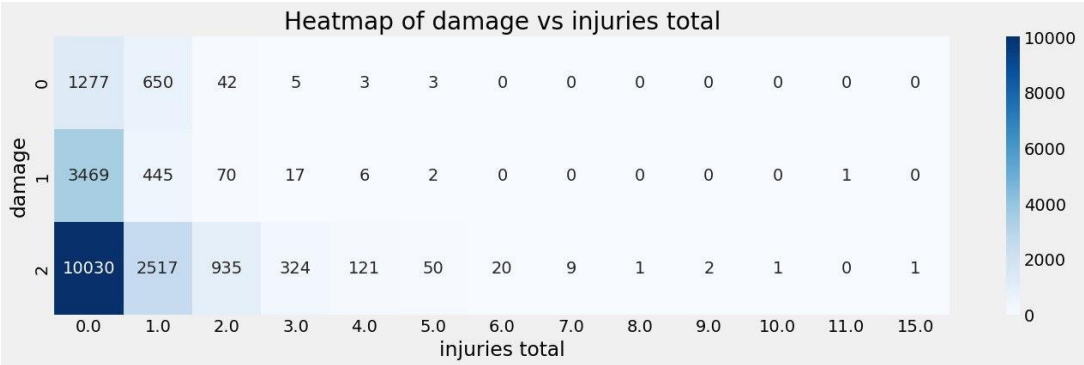
- The partial heatmaps:

We now use partial heatmaps to illustrate the relationship between damage severity and other variables, where the colors in these plots represent the density of the relationship between the two variables.

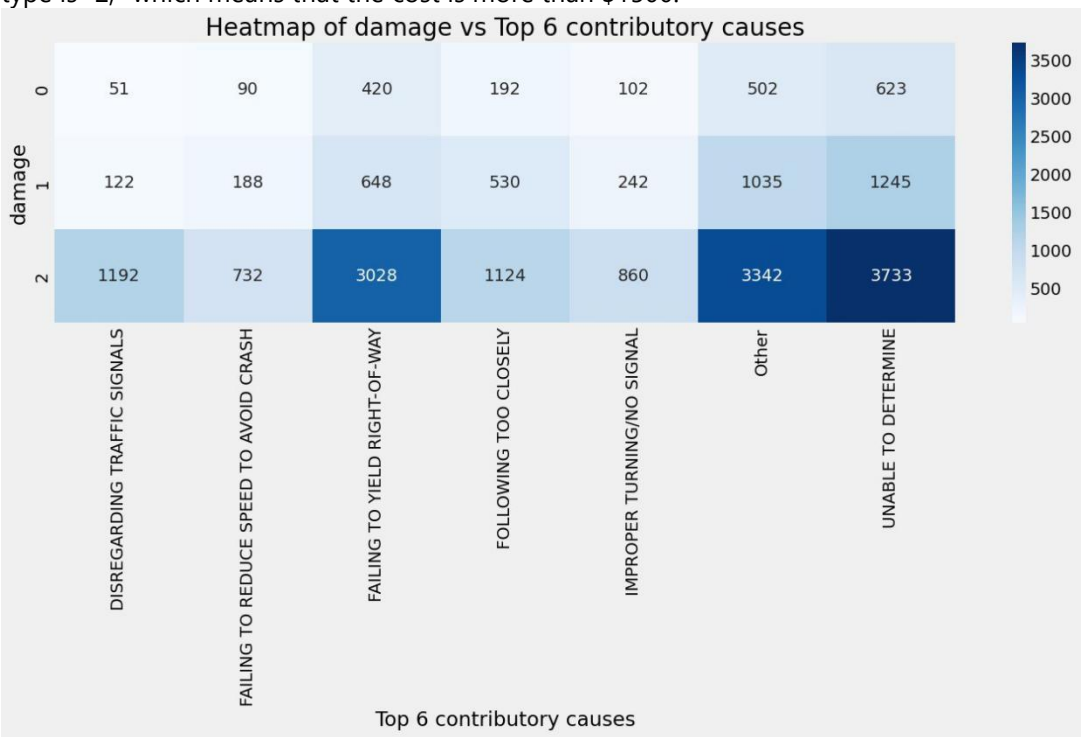


The relationship between damage severity and the time of the incident (damage vs. time_of_day):
The plot shows the distribution of accidents across different times of the day and their impact on damage severity. It can be seen that most high-cost injuries occur during “time 2,” which

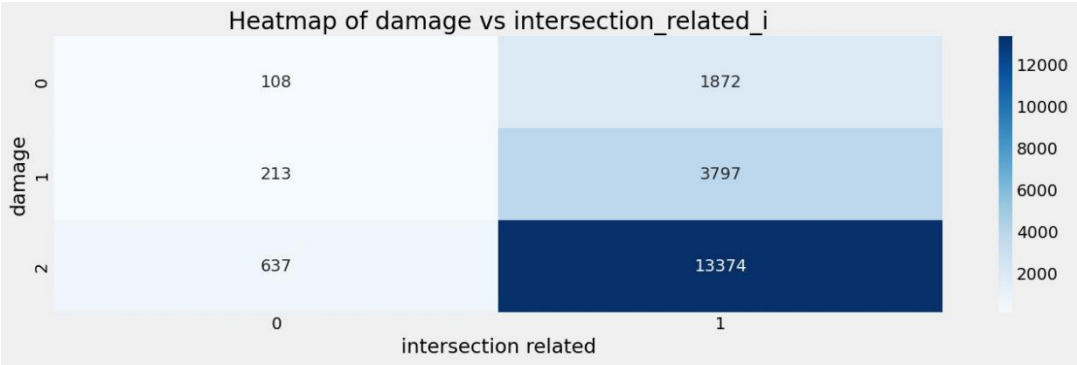
refers to the “evening”. This might be due to blurred vision or traffic congestion in the evening.



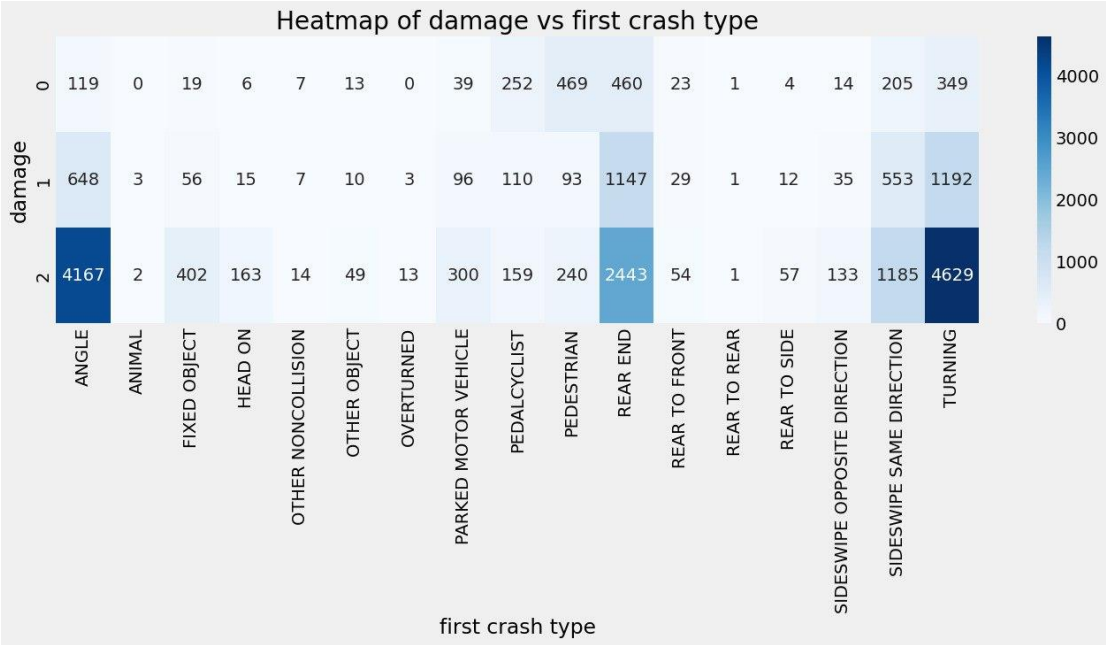
The relationship between damage severity and the number of injuries (damage vs. injuries_total):
The plot shows the distribution of injuries across different levels of damage. The higher the number of injuries, the more likely the damage is severe.
It can be observed that incidents with more than 5 injuries often indicate that the damage type is “2,” which means that the cost is more than $1500.



The relationship between damage severity and the top contributing causes (damage vs. Top_6_contributory_causes):
The plot shows the most common contributing factors leading to severe accidents, such as speeding and disregarding traffic signals. It can be noted that the most common and perhaps most impactful cause of severe damage is “Failing to yield right of way,” followed by “Disregarding traffic signals.”



The relationship between damage severity and the location of the incident (damage vs. intersection_related_i):
The plot shows how accidents occurring at intersections correlate with damage severity.
It is clearly shown that intersections often experience more severe accidents due to the presence of more vehicles and pedestrians.



The relationship between damage severity and the type of first crash (damage vs. first_crash_type):
The plot shows how the type of first crash (e.g., head-on, side-impact, pedestrian, etc.) affects the level of damage.
It can be seen that some types of collisions are more strongly associated with severe damage, such as the “angle” and “Rear end” types, helping to identify the most dangerous accidents.

5. Modeling Process:

a. Data Splitting and Model Application:

Initially, the dataset was split into two parts: a training set (80%) and a testing set (20%) using the train_test_split function. This split ensures that the models are trained on one part of the data and tested on another, allowing for an accurate evaluation of model performance.

Applying the K-Nearest Neighbors (KNN) Model
The KNN model with 7 neighbors was used to train the data. After training, the model was applied to the test set to predict the classes. The results showed that the model's overall accuracy was 67%, with a significant performance in classifying category 2 (representing the majority of accidents), achieving an accuracy of 89%. However, the accuracy for predicting categories 0 and 1 was noticeably lower, reflecting the model's difficulty in handling less frequent categories.

Applying the Logistic Regression Model
Next, the Logistic Regression model was applied, showing a higher overall accuracy compared to the KNN model, with an accuracy of 71%. This model performed well in classifying category 2, with a marked improvement in predicting categories 0 and 1 compared to the KNN model. However, the accuracy in predicting category 1 was still less than ideal, as category 2 was sometimes predicted instead of category 1.

b. Features Selected and The Hyperparameters Chosen:

A set of important features has been chosen to help classify accidents with higher accuracy:
- 'intersection_related_i'
- 'most_severe_injury'
- 'injuries_total'
- 'crash_hour'
- 'crash_day_of_week'
- 'crash_month'
- 'time_of_day'
- 'num_units'

Selected Hyperparameters In the KNN model, the number of neighbors (k) was set to 7 based on previous experiments. This choice helped improve performance by balancing classification accuracy and computational efficiency.

In the Logistic Regression model, max_iter was set to 1000 to increase the number of iterations and ensure model stability.

6. Evaluation:

Logistic regression outperformed KNN in overall accuracy, achieving 71% compared to 67% for KNN, making it more reliable in predictions. For the most severe damage category (Class 2), both models performed similarly. However, for the medium damage category (Class 1), logistic regression was clearly superior, while KNN struggled to classify it correctly.

In terms of balance, logistic regression was more stable, whereas KNN tended to classify most samples into the most frequent category, reducing its accuracy for less common classes. Additionally, logistic regression is faster and more computationally efficient since it relies on direct calculations, whereas KNN requires computing distances between all points, making it slower during predictions.

Conclusion: Logistic regression is the better choice due to its higher accuracy, balanced performance across categories, and faster computations, making it more efficient than KNN for this classification task.

7. Conclusions and Insights:

This project demonstrates that analyzing accident data using machine learning can help predict damage severity and identify the factors influencing it, contributing to improved traffic safety measures and more informed decision-making. The results indicate that the number of injuries, time of the accident, type of collision, and location play a crucial role in determining damage levels, with accidents occurring in the evening and at intersections being particularly severe. The study also revealed that failing to yield the right of way and disregarding traffic signals are among the most common causes of high-damage accidents. When comparing models, logistic regression outperformed KNN in terms of accuracy and its ability to classify damage levels more consistently.

Based on these findings, predictions can be enhanced by incorporating additional data, such as road conditions and driver behavior, and exploring more advanced machine learning models. Furthermore, these insights can be leveraged to strengthen traffic monitoring and improve road design to reduce accident risks. This project highlights that machine learning can serve as a powerful tool in developing intelligent solutions to minimize accidents and mitigate their impact.